# Dylan Alexander Krishnan

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Hands-on experience in securing Linux infrastructure, deploying SIEM-based detection tools, and performing advanced in-depth malware analysis. Proven ability to design network architecture that is scalable & secure. Strong Python skills used for automation, data processing, and workflow optimization, complemented by knowledge of threat detection and system hardening.

## Education

- The Pennsylvania State University, University Park, PA | May 2026
- B.S. Cybersecurity Analytics and Operations | GPA: 3.7

## Skills

- **Security Tools & Frameworks:** Keycloak (SSO/IAM), Active Directory, OPNsense, pfSense, Wazuh, Ghidra, Postman (API Security Analysis), Tailscale (ZTNA), IDS/IPS, Docker, Network Segmentation, RBAC, Zero Trust, Defense-in-Depth
- **Programming & Scripting:** Python (Automation, API Integration, Reverse Engineering, Algorithm Development), SQL, Bash, Selenium
- **Platforms & Infrastructure:** Linux (Debian, RHEL, Ubuntu, Rocky, Arch), Proxmox VE, KVM, Xen, Hyper-V, Windows, Cloud Platforms

## Security & Infrastructure Projects

**Security-Focused Infrastructure Homelab | Proxmox VE, OPNsense, Rocky Linux**

- Designed and maintained a production-style homelab environment running Proxmox VE hypervisor (based on Debian Linux) on HP ProLiant Gen9 hardware
- Deployed and hardened self-hosted services including:
  - Bitwarden for secure password management with role-based access controls
  - Nextcloud for encrypted file storage, calendar, and task management
  - Self-hosted Git instance for secure version control and development workflows
  - Deployed DroneCI as a self-hosted CI/CD tool for Git automation and deployment
  - Keycloak for centralized Single Sign-On (SSO) and identity management across services
- Implemented network segmentation and firewall policies using OPNsense to isolate services and minimize attack surface
- Configured secure reverse proxying with Cloudflare and secure access with Tailscale

**Isolated Malware Analysis Environment Architecture | Proxmox VE, OPNsense**

- Designed and implemented a fully isolated virtualized network environment for safe malware analysis using a virtualized OPNsense firewall, following zero-trust and defense-in-depth principles
- Engineered network segmentation to prevent malware traffic from reaching host or private networks using a hardened firewall-based architecture
- Implemented Suricata IDS/IPS system to automatically detect and block malicious traffic from reaching other networks

**SOC Forensics & Ransomware Incident Investigation | Volatility 3, Wireshark, Autopsy**

- Performed multi-source forensic investigation across 4 compromised Windows workstations, combining disk imaging, memory analysis, and network traffic analysis
- Utilized Wireshark and disk image analysis to identify full kill chain from WinRM lateral movement and ~270 MB pre-encryption data
- Recovered AES decryption key and IV from live ransomware process memory dump using Volatility 3, enabling full file recovery without paying a $2.5M Bitcoin ransom

**API Security Analysis & Reverse Engineering Project | Python, Linux**

- Analyzed undocumented internal APIs to understand request structures for automation
- Demonstrated how reverse engineering techniques can be used to understand an API using Postman
- Uncovered critical access control vulnerabilities including excessive data exposure and improper authorization validation

**AI Data Mining Automation Tool | Python, OpenAI API**

- Developed AI-powered data mining tool that extracts scholarship requirements from web HTML and evaluates candidate eligibility using structured inputs and AI natural language processing
- Integrated the OpenAI API to analyze unstructured scholarship requirements and apply consistent eligibility logic

**Website Scraping and Automation (Proof of Concept) | Python, Selenium**

- Engineered a web automation bot to navigate application workflows without human intervention
- Exposed how unattended automated agents can repeatedly probe and interact with time-sensitive, access-controlled web resources

## Professional Experience

**Systems Engineer – Countryside Broadband** | Remote | *January 2025 – Present*

- Architected and hardened a multi-hypervisor virtualized environment (Xen, KVM) to enforce workload isolation and contain lateral movement risk across production infrastructure
- Engineered stateful firewall ruleset and BGP routing policies in pfSense to enforce strict ingress/egress traffic control across the organization's network perimeter
- Reverse engineered undocumented internal API endpoints using Python to automate account provisioning, eliminating manual processes and reducing human error in access management workflows
- Deployed and managed endpoint security and patch management via Action1 across all employee devices, maintaining a hardened, up-to-date fleet

**Software Engineer Intern – Pfizer Commercial Analytics** | New York, NY | *June 2025 – August 2025*

- Engineered least-privilege SQL queries to scope sensitive commercial data access strictly to authorized executive reporting requirements in a regulated pharmaceutical environment
- Enforced secure development practices using GitHub version control and Jira audit trails across all data pipeline work involving confidential commercial data

**Frontend Dev Intern – Pfizer R&D** | Groton, CT | *May 2023 – August 2023*

- Replaced an insecure, email-based outsourcing workflow by engineering a centralized platform with enforced audit trails and documented approval chains. Deployed to 2,000+ internal users
- Reduced outsourcing turnaround time by 75% while simultaneously improving compliance posture and accountability across all cross-departmental legal requests

## Additional Experience

**Embedded Custom Smart TV OS Project | Arch Linux, Embedded Systems**

- Built and hardened a custom Arch Linux-based Smart TV OS optimized for home theater, deploying the YouTube TV web app via Chromium Embedded Framework
- Enabled 1080p Netflix streaming through Widevine DRM configuration and integrated native Apple AirPlay support for seamless screen mirroring